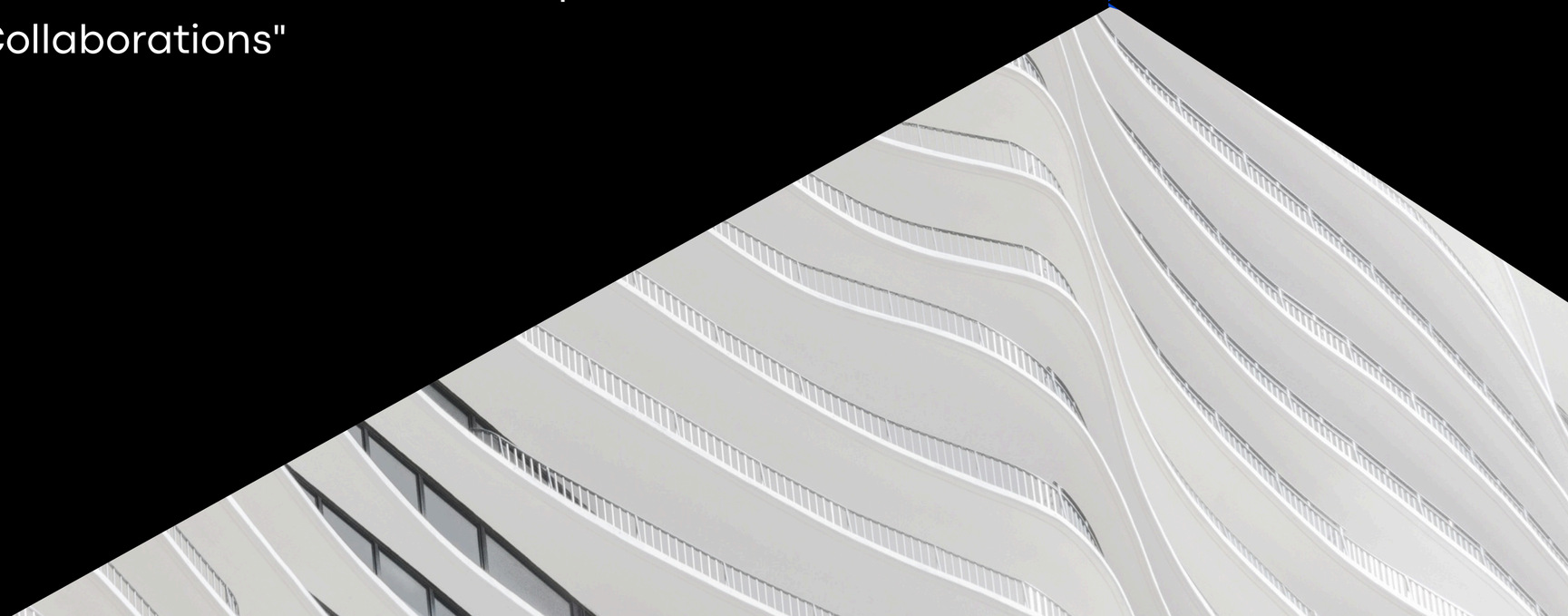
CodeSphere-Ledger

A Global Blockchain-Powered Hub for Secure Developer-Client Collaborations"



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PROBLEM

- UnSecured Financial Transactions on Freelance platforms
- High Transaction Fees on Freelance Platforms
- Delayed Payments
- Lack of Transparency

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Current Scenario

Finding work: 66% of freelancers say finding enough work is a challenge.

Late payments: 58% of freelancers have experienced late payments or not getting paid at all

About 20-25% of users have concerns over data privacy or have experienced breaches, leading to distrust of the platform.

Around 35-45% of freelancers and clients have experienced delayed or unsatisfactory dispute resolution on traditional platforms.

Proposed Solution

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1. **Code** : Represents the platform's focus on software development, highlighting its user base of developers who create and innovate.
2. **Sphere** : Suggests an inclusive ecosystem where clients and developers collaborate, foster community or project execution
3. **Ledger** : Refers to the use of blockchain technology for secure, transparent transaction recording, ensuring trust and accountability between users.

Together, these elements convey a transformative freelance platform that enhances connectivity, transparency, and efficiency in the development marketplace.

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Tech Stack

Programming Languages:
TypeScript,Python

Frameworks/Libraries:
NextJS,React.js, Kafka
Scikit-Learn,Pandas,Pickle

BlockChain:
Solana, Solana Wallet,
SmartContract, Rust

Deployment:
GitHub,Docker(CI/CD)

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Methodology

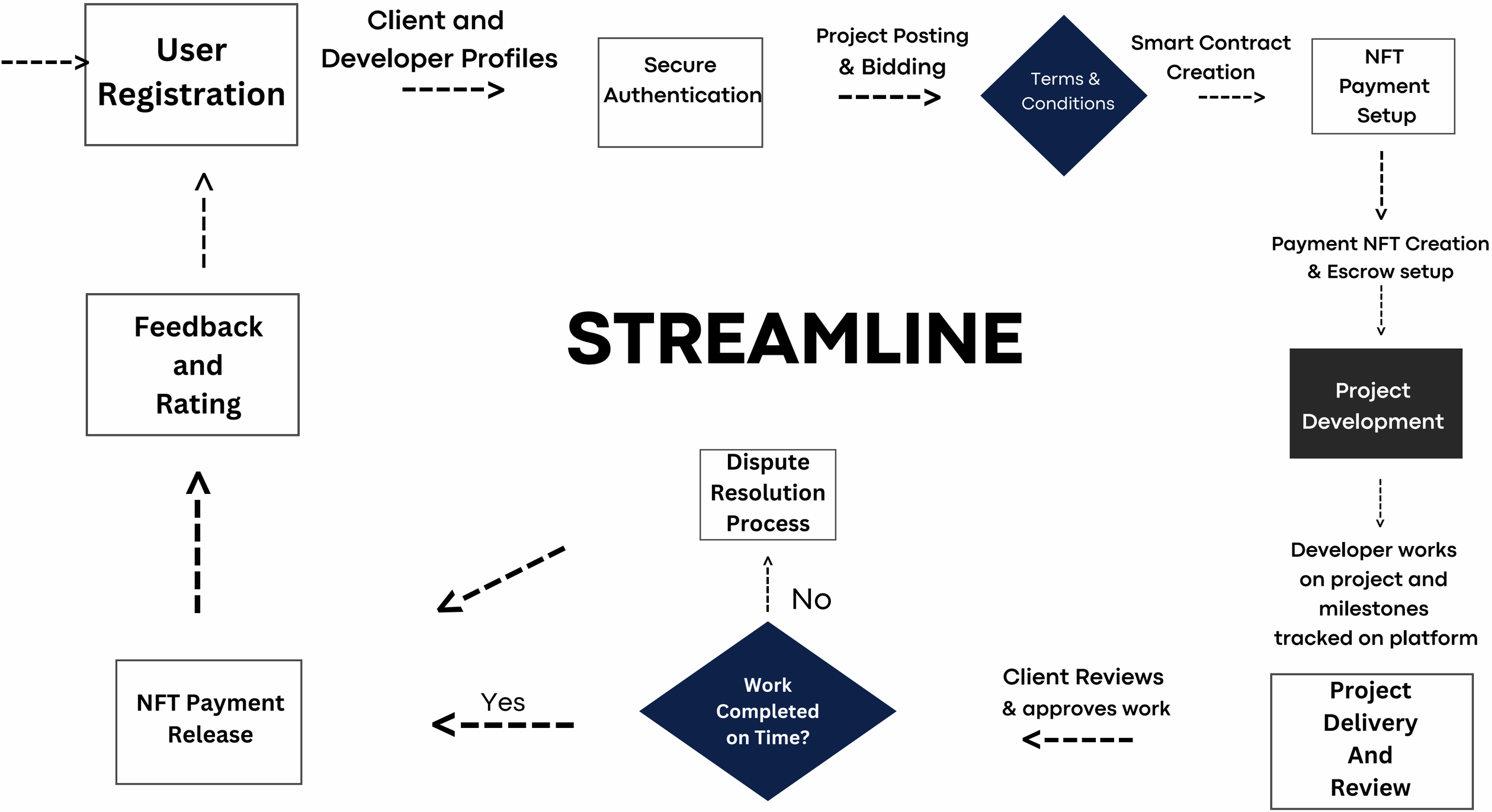
- **Decentralized Marketplace**: The platform uses blockchain to connect clients and developers, ensuring secure, transparent interactions without intermediaries.

- **Smart Contracts**: Automated smart contracts handle payments and agreements, guaranteeing instant, dispute-free transactions upon project completion.

- **Machine Learning Matching**: A machine learning algorithm matches clients with developers based on project requirements and skills, optimizing project success.

- **NFTs for Ownership**: NFTs ensure verifiable ownership of completed work, enabling secure transfer of intellectual property.

STREAMLINE


User Registration
Client and Developer Profiles
Secure Authentication
Project Posting & Bidding
Terms & Conditions
Smart Contract Creation
NFT Payment Setup
Payment NFT Creation & Escrow setup
Project Development
Developer works on project and milestones tracked on platform
Project Delivery And Review
Client Reviews & approves work
Work Completed on Time?
No
Dispute Resolution Process
Yes
NFT Payment Release
Feedback and Rating

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Current Progress

Problem Statement Validation:

The core issues for our project have been deeply analyzed, with clear problem statement driving each solution. We've aligned the problem-solving approach with real-world use cases and scalability.

Tech Stack Finalization:

The tech stack for our project is finalized based on performance, scalability, and integration needs. Integration with smart contract and decentralized identity for seamless, secure transactions is underway.

Platform Architecture and Blueprint:

We've created a structured blueprint for the entire platform workflow, ensuring optimal data flow, security, and scalability.

Prototype and Business Model Development :

The business model is also defined, outlining revenue streams (e.g., ad-based revenue, premium services) and market entry strategies.

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Future Scope

Freemium + Premium Model:
Allow basic access for free to developers and clients but introduce premium features

Ad-Based Revenue:
Allow companies to advertise developer tools, software, or other services on the platform, creating an additional revenue stream.

Dynamic Rating System:
Contest performance will generate ratings, helping clients easily identify top talent.

Smart Contract Advancement :
Using ReInforcement Learning To Include the terms due to which Dispute creates in History.

Partnerships & Collabration :
We plan to introduce partnership programs where top developers can collaborate with others to offer combined services or specialized packages, enabling team formation and driving larger-scale projects on the platform.

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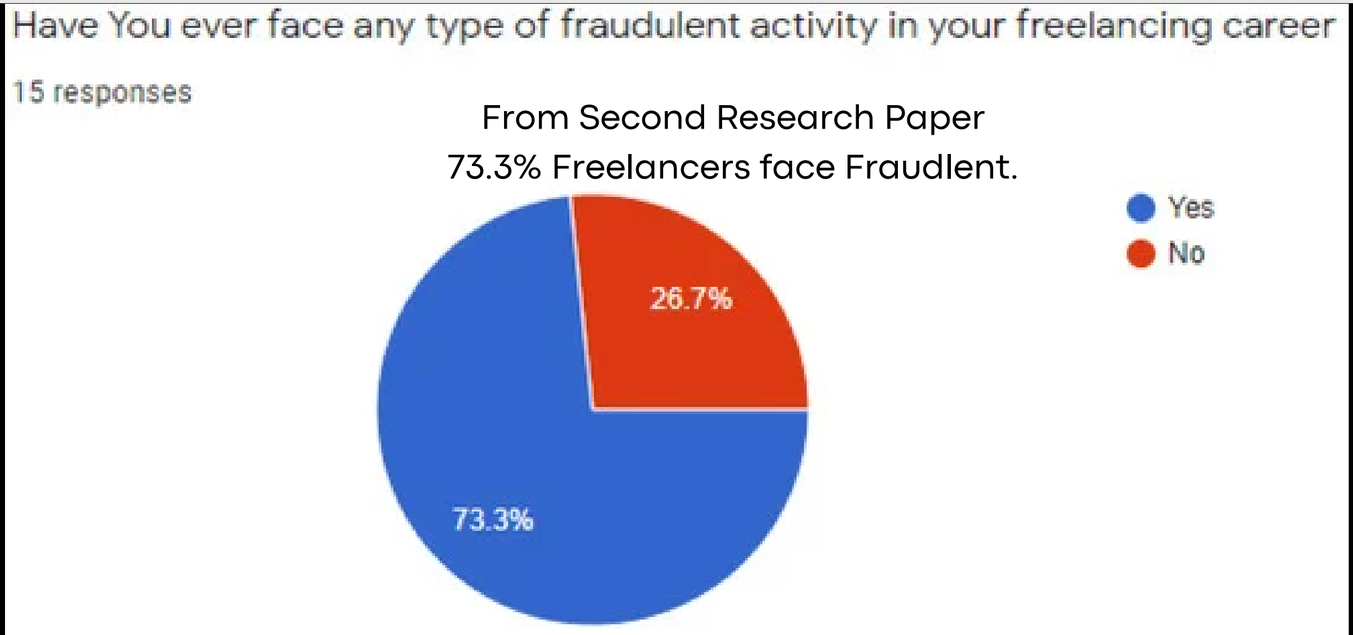
References

Research Paper From Google Scholar describing problem based on surveys.

- https://ieeexplore.ieee.org/abstract/document/9182127
- https://www.mdpi.com/2079-9292/11/5/789



From Second Research Paper
73.3% Freelancers face Fraudlent.

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-Thank You